Today

Populations vs. Samples (quick)

Statistics and Sample Spaces

Distributions

Quantiles

Distribution Shapes

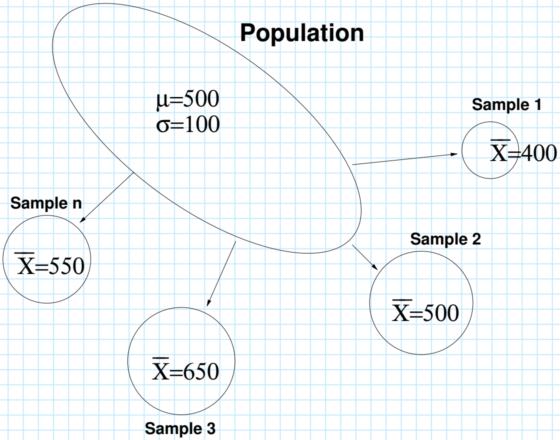
Central Tendency

Populations vs. Samples

- Populations $\rightarrow$ Parameters
- Samples $\rightarrow$ Statistics

	Statistic	Parameter
Mean	$\overline{X}$	μ
Variance	s^2	σ^2
Correlation	r	ρ

Populations vs. Samples



Descriptive vs. Inferential Statistics

- Descriptive statistics: describe the characteristics of a group or population
- Inferential statistics: infer characteristics of a population based characteristics of a sample from that population.

What is a statistic?

A statistic is any number (or numbers) mathematically derived from a data set. It is important that you think of a statistic as a measurement of a dataset, as opposed to the measurements we obtain for an individual in that dataset.

Statistics

Definitions (Sample and Statistic)

A *sample* is a collection of measurements, often written as

$$X = \{X_1, X_2, X_3, \ldots, X_N\},$$

where N is the number of measured units (individuals) in the sample. A *statistic* $S(X)$ is any quantity or quantities computed from X:

$$S(X) = f(X_1, X_2, \ldots, X_N).$$

Example

$$\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$$

Variables and Sample Spaces

Sample Space or Support

- A discrete variable X_i can take on values from a finite or countably infinite set $\{x_1, x_2, \ldots\}$.
- A continuous variable X_i can take on values in a range like $[0, 1]$ or $[0, \infty)$.
- X_i is a placeholder for individual i's measurement and x_i is the observed value for that measurement.

Load the Data

```
data <- read.csv("GSS_1980.csv",header=TRUE)
head(data)

# Delete the respondent ID
data$X <- NULL
head(data)

# Give the variables in the dataframe nice names (be careful about order)
names(data) <- c("income","sibs","age","happy","health","taught")

# Compute the frequency and relative frequency distributions
table(data$health)
table(data$health)/length(data$health)
```

Output

```
> data <- read.csv("GSS_1980.csv",header=TRUE)
>
> # Take a peek at the first few lines of the dataset:
> head(data)
      X REALRINC SIBS AGE HAPPY HEALTH YOUNGEN
1 10653     8308    4  62     3      2       1
2 10654    11329   15  77     1      2       2
3 10656    16994    0  28     1      2       2
4 10657   139297    5  24     2      1       1
5 10659    32100    4  63     2      1       1
6 10661    56647    1  62     1      2       2
> # We don't need the respondent ID number so we'll delete it:
> data$X <- NULL
> head(data)
  REALRINC SIBS AGE HAPPY HEALTH YOUNGEN
1     8308    4  62     3      2       1
2    11329   15  77     1      2       2
3    16994    0  28     1      2       2
4   139297    5  24     2      1       1
5    32100    4  63     2      1       1
6    56647    1  62     1      2       2
> # Give the variables in the dataframe nice names (be careful about order)
> names(data) <- c("income","sibs","age","happy","health","taught")
> # Note that the $ character lets you pull out a variable from a data frame:
> table(data$health) # frequency distribution
  1   2   3   4 
319 400 132  22 
> table(data$health)/length(data$health) # relative freq distribution
         1          2          3          4 
0.36540664 0.45819015 0.15120275 0.02520046 
```

GSS “Health” Question

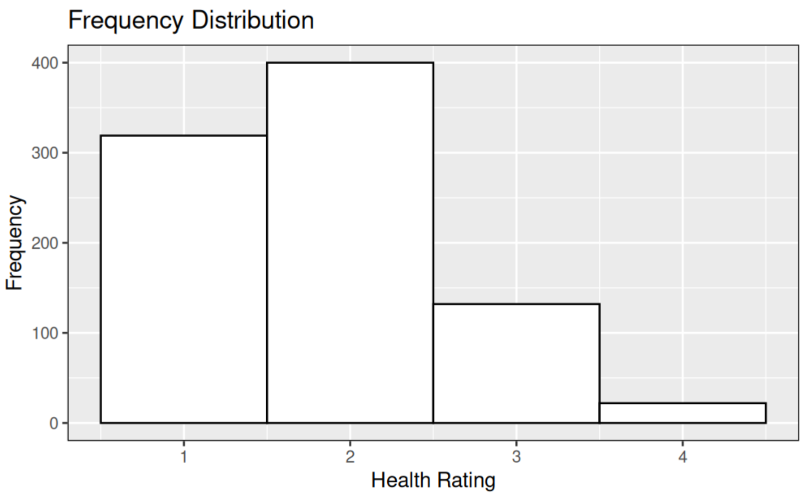
Sample space $S = \{1, 2, 3, 4\}$

	Health Rating			
Raw Frequency	1	2	3	4
$f_X(x)$	319	400	132	22

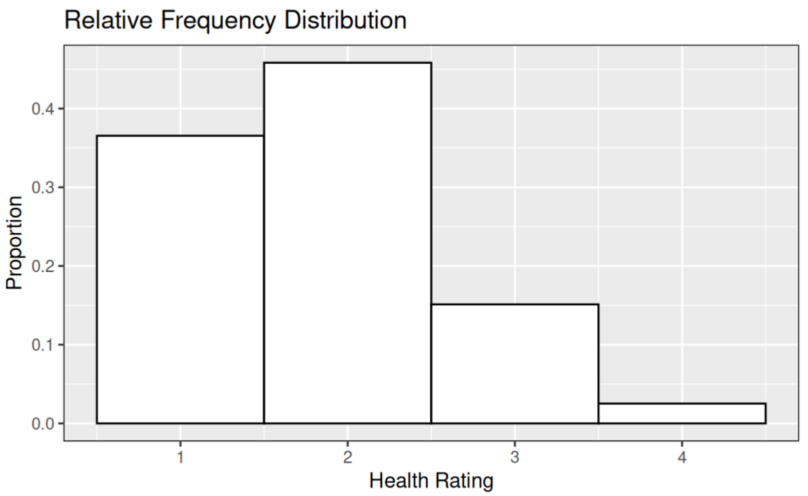
	Health Rating			
Relative Frequency	1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03

What is $Pr(X = 1)$?

(Raw) Frequency Distribution



Relative Frequency Distribution



Cumulative Distributions

The number/proportion of observations equal to or below x_i.

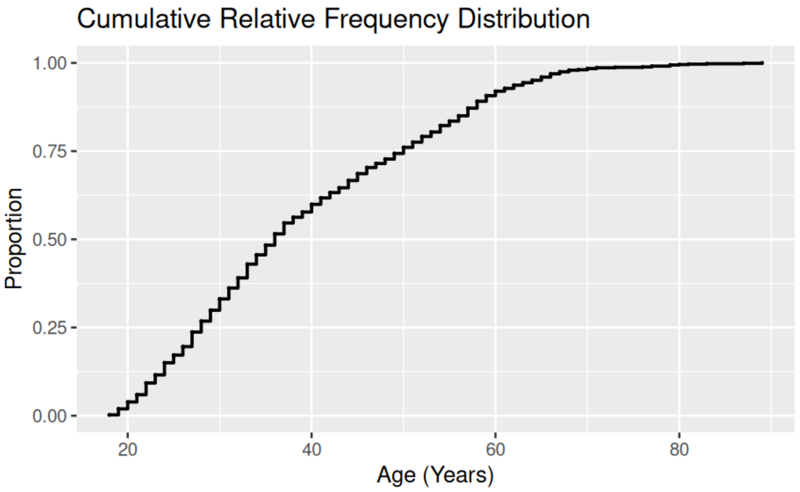
$$\begin{aligned}\text{(Frequency) } F_X(x_i) &= \sum_{y \le x_i} f_X(y) \\ \text{(Relative Frequency) } P_X(x_i) &= \sum_{y \le x_i} p_X(y)\end{aligned}$$

$P_X(x) \equiv Pr(X \le x)$

GSS "Health" Question

	Health Rating			
Relative Frequency	1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03
$P_X(x)$				

Respondent Age



Quantiles

A set of quantiles are the cutpoints that divide a distribution into intervals with equal proportions.

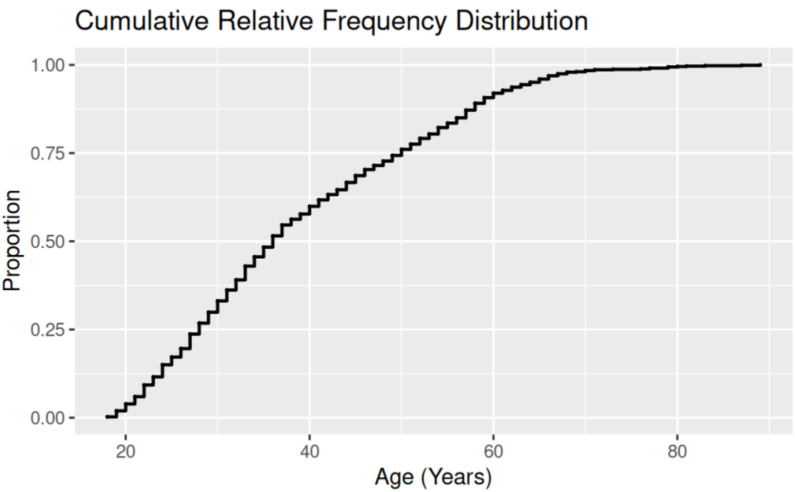
(1) Median (50%)
(2) Terciles (33%)
(3) Quartiles(25%)
(4) Quintiles (20&)
(9) Deciles (10%)
(19) Ventiles (5%)
(99) Percentiles (1%)

```
# Compute quantiles
median(data$age)
quantile(data$age,seq(0,1,by=.05)) # Vary "by=" for other quantiles
```

Respondent Age

Estimate the

- median
- quartiles
- 45^{th} percentile

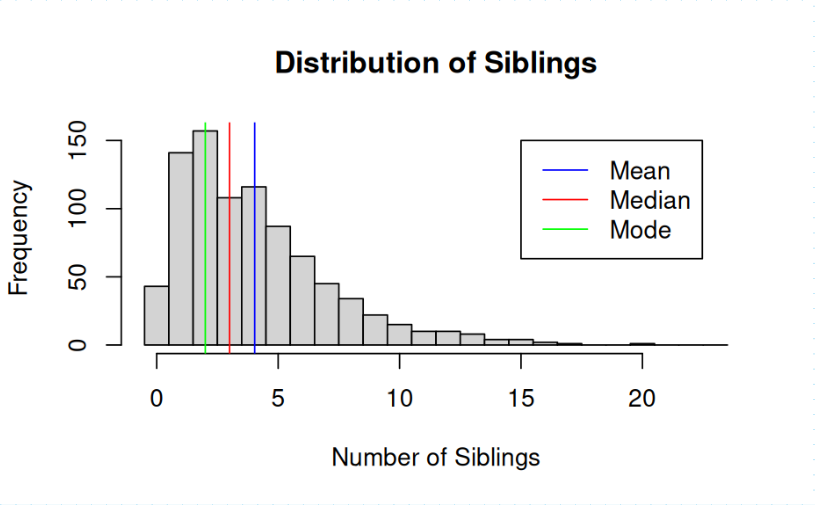


- What proportion of the respondents are younger than 45?
- Percentile vs. Percentile Rank

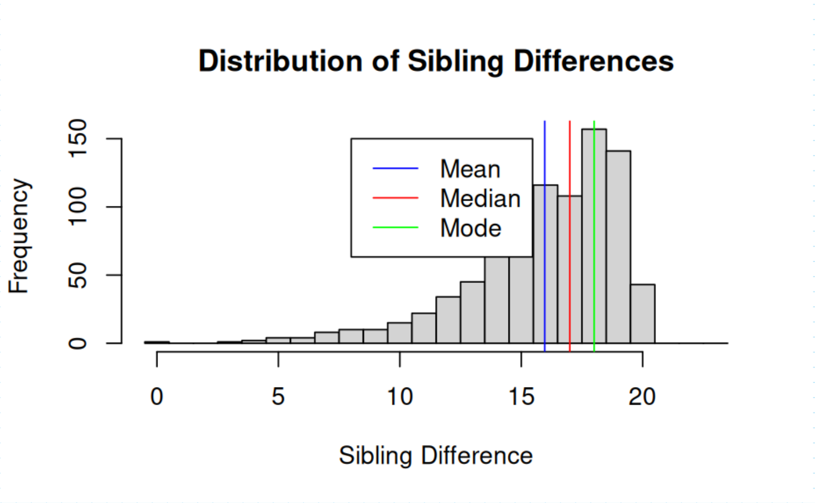
Distribution Shape

- Skewness
- Symmetry
- Modes

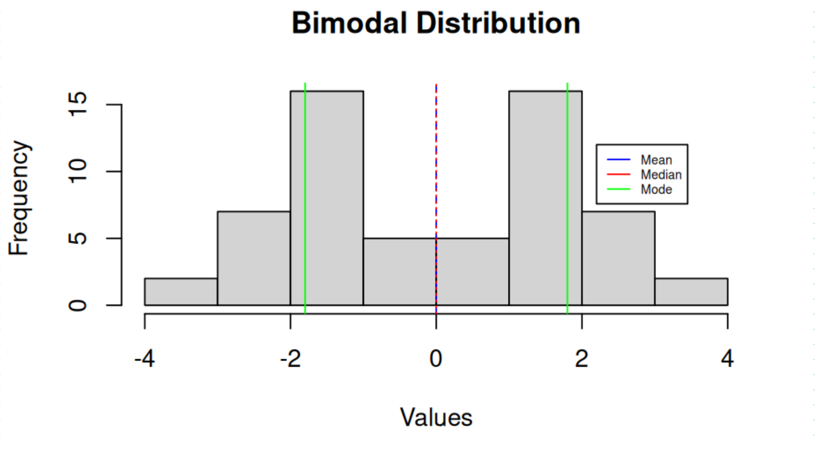
Positively Skewed



Negatively Skewed



Symmetric and Bimodal



Asymmetric and Bimodal

Multimodal

Why Do We Care?

The shape of a distribution determines:

- what kinds of statistics are most appropriate to describe it.
- what kinds of statistics are most appropriate for inference.
- what kinds of statistical *models* we should use for explanation and prediction.

Measures of Central Tendency

The middle or most characteristic score; a measurement of the dataset X.

- mean: $\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$.
- median: If $Pr(X \leq x) \geq .5$ and $Pr(X \geq x) \geq .5$, then $Md = x$
- mode: If $x = \text{argmax}_{x_i} Pr(X = x_i)$, then $Mo = x$.

GSS "Health" Responses

Relative Frequency	Health Rating 1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03
$P_X(x)$	0.37	0.83	0.98	1.00

- mean
- median
- mode

Weighted Means and Expected Values:

$$\overline{X} = \sum_{i=1}^{m} x_i p_X(x_i)$$

Weighted Means

```
> # Central tendency
> library('DescTools') # Has some useful stuff
> mean(data$health)
[1] 1.836197
> median(data$health)
[1] 2
> Mode(data$health) # Mode function from DescTools package)
[1] 2
attr(,"freq")
[1] 400
>
> # Weighted mean
> p.Health <- table(data$health)
> Health <- as.numeric(names(p.Health))
```